

Graphs of functions - Set 1 - Solutions

Specification MM8.1, MM8.5-MM8.7, M4.15.

Answer key

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Worked solutions

Question 1 GRA-01 · Graphs of functions · spec MM8.6

How many times does the graph of $y = x^3 - x^2 - 6x$ cross or touch the x -axis?

Answer: A

$x(x^2 - x - 6) = x(x - 3)(x + 2)$; three distinct real roots, so three crossings.

Full method - how and why

Axis crossings are roots. Factorise fully and count **distinct** roots.

- 1 Factor out x : $x^3 - x^2 - 6x = x(x^2 - x - 6)$.

why: There is no constant term, so x is a factor - always check this first.

- 2 Factorise the quadratic: $x^2 - x - 6 = (x - 3)(x + 2)$.

why: Two numbers multiplying to -6 , adding to -1 .

- 3 Roots: $x = 0, 3, -2$ - three distinct values, each a simple root, so the graph crosses three times.

why: Distinct simple roots mean genuine crossings; a repeated root would be a touch.

Where it goes wrong

- Missing the root $x = 0$ and answering 2.
- Counting the y -intercept as an extra crossing of the x -axis.

Takeaway Factor out x when there is no constant term; count distinct roots for the number of axis meetings.

Question 2 GRA-02 · Graphs of functions · spec MM8.7

The graphs of $y = x^2 - 2x$ and $y = k$ meet at exactly two points. Find the complete set of values of k .

Answer: A

$y = (x - 1)^2 - 1$ has minimum -1 ; a horizontal line above the minimum meets the parabola twice.

Full method - how and why

A horizontal line meeting a parabola twice: the line must be above the minimum.

- 1 Complete the square: $x^2 - 2x = (x - 1)^2 - 1$, so the minimum value is -1 at $x = 1$.

why: The turning point governs how many times a horizontal line can cut the curve.

- 2 A line $y = k$ meets the parabola twice when k is **strictly above** the minimum.

why: At $k = -1$ the line is tangent at the vertex: exactly one point. Below it: none.

- 3 So $k > -1$.

why: Strict inequality, because equality gives only one intersection.

Where it goes wrong

- Including $k = -1$ and choosing $k \geq -1$ - offered as a distractor.
- Using the y -intercept 0 instead of the minimum value.

Takeaway Count intersections with a horizontal line by sliding it against the turning point; equality gives tangency.

Question 3 GRA-03 · Graphs of functions · spec MM8.1

Which of the following could be the equation of a cubic curve that touches the x -axis at $x = 2$ and crosses it at $x = -1$, and has a negative y -intercept?

Answer: A

Touching at 2 needs the factor $(x - 2)^2$ and crossing at -1 the factor $(x + 1)$. $y = (x - 2)^2(x + 1)$ has y -intercept $4 > 0$, so the sign is reversed: $y = -(x - 2)^2(x + 1)$ with intercept -4 .

Full method - how and why

Build the cubic from its root behaviour, then use the y -intercept to fix the sign.

- 1 'Touches at $x = 2$ ' means a repeated factor $(x - 2)^2$; 'crosses at $x = -1$ ' means a simple factor $(x + 1)$.

why: Even-power factors touch the axis; odd-power factors cross it.

- 2 So the cubic is $y = a(x - 2)^2(x + 1)$ for some constant $a \neq 0$.

why: Degree $2 + 1 = 3$ ✓.

- 3 Compute the y -intercept: at $x = 0$, $y = a(4)(1) = 4a$.

why: Substitute $x = 0$ into the factorised form.

- 4 A negative intercept needs $a < 0$, so $y = -(x - 2)^2(x + 1)$ works (intercept -4).

why: The option with $a = +1$ has intercept $+4$ and is the trap.

Where it goes wrong

- Choosing $(x - 2)^2(x + 1)$ and ignoring the sign of the intercept.
- Swapping which root is repeated, e.g. $(x - 2)(x + 1)^2$.

Takeaway Repeated factor $\Rightarrow$ touch; simple factor $\Rightarrow$ cross; the y -intercept fixes the leading sign.

Question 4 GRA-05 · Graphs of functions · spec MM8.1

Which of the following statements about the graph of $y = |2x - 4|$ is/are true?

- I It meets the x -axis at exactly one point.
- II It is symmetric about the line $x = 2$.
- III It lies entirely on or above the line $y = 0$.

Answer: H

$|2x - 4| = 0$ only at $x = 2$ (I). $|2x - 4| = 2|x - 2|$ is symmetric about $x = 2$ (II). A modulus is never negative (III).

Full method - how and why

Three properties of a modulus graph. Rewriting $|2x - 4|$ as $2|x - 2|$ makes all three obvious.

- 1 Rewrite: $|2x - 4| = 2|x - 2|$.

why: Factoring the 2 out of the modulus is legitimate because $|2u| = 2|u|$.

- 2 I: the graph meets $y = 0$ when $|x - 2| = 0$, i.e. only at $x = 2$. **True.**

why: A V-shape touches the axis at its single vertex.

- 3 II: $|x - 2|$ depends only on the distance from 2, so the graph is symmetric about $x = 2$. **True.**

why: Replacing x by $4 - x$ leaves the value unchanged.

- 4 III: a modulus is never negative, so the graph lies on or above the x -axis. **True.**

why: $|u| \geq 0$ by definition.

- 5 All three.

why: Match to the option list.

Where it goes wrong

- Thinking the graph crosses the axis twice because $|x^2 - 4|$ -type graphs do - this one is linear inside the modulus.
- Reading the line of symmetry as $x = 4$.

Takeaway $|ax + b| = |a||x + \frac{b}{a}|$: a V-shape with its vertex at the root, never below the axis.

Question 5 GRA-06 · Graphs of functions · spec MM8.6

The curve $y = x^4 - 5x^2 + 4$ meets the coordinate axes at how many points in total?

Answer: A

x -axis: $(x^2 - 1)(x^2 - 4) = 0$, four points. y -axis: $(0, 4)$, one point. Total 5.

Full method - how and why

Count both axes. The x -axis meetings come from the roots; the y -axis always contributes exactly one point.

1 Factorise: $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$.

why: Quadratic in x^2 : two numbers multiplying to 4 and adding to -5 .

2 Solve: $x = \pm 1$ and $x = \pm 2$ - four x -axis crossings.

why: Each positive value of x^2 gives two values of x .

3 y -axis: substitute $x = 0$ to get $y = 4$ - one point.

why: A function meets the y -axis exactly once.

4 Total: $4 + 1 = 5$.

why: The y -intercept is a separate point (it is not one of the roots here, since $y(0) \neq 0$).

Where it goes wrong

- Forgetting the y -axis and answering 4.
- Counting $x^2 = 1, 4$ as only two points.

Takeaway 'Meets the coordinate axes' means both axes: roots plus the single y -intercept.

Question 6 GRA-09 · Graphs of functions · spec MM8.1

The graph of $y = \log_2 x$ is reflected in the line $y = x$. The resulting graph has equation

Answer: A

Reflection in $y = x$ gives the inverse function: $x = \log_2 y$, i.e. $y = 2^x$.

Full method - how and why

Reflection in $y = x$ produces the inverse function - swap x and y and re-solve.

1 Reflecting in $y = x$ swaps the roles of x and y : the new graph satisfies $x = \log_2 y$.

why: Every point (a, b) maps to (b, a) .

2 Rewrite in exponential form: $y = 2^x$.

why: $\log_2 y = x$ means $y = 2^x$ - logs and exponentials are inverse functions.

Where it goes wrong

- Choosing $y = \log_2(-x)$ (a y -axis reflection) or $-\log_2 x$ (an x -axis reflection).
- Thinking the inverse of a log is x^2 because of the base 2.

Takeaway Reflection in $y = x$ gives the inverse; $y = a^x$ and $y = \log_a x$ are mirror images in that line.

Question 7 GRA-10 · Graphs of functions · spec MM8.6

A polynomial of degree 5 with real coefficients has how many real roots at least, and at most?

Answer: A

An odd-degree polynomial tends to opposite infinities at each end, so it crosses the axis at least once; a degree-5 polynomial has at most 5 roots.

Full method - how and why

Two standard facts about polynomial roots: the maximum is the degree, and odd degree forces at least one real root.

- 1 At most: a degree-5 polynomial has at most 5 roots, since each root contributes a linear factor.
why: More than 5 factors would raise the degree above 5.
- 2 At least: as $x \rightarrow +\infty$ and $x \rightarrow -\infty$, an odd-degree polynomial tends to opposite infinities.
why: The leading term ax^5 dominates and changes sign with x .
- 3 A continuous function that is positive somewhere and negative somewhere must cross zero, so there is at least one real root.
why: Intermediate value reasoning; this is why every odd-degree polynomial has a real root.
- 4 At least 1, at most 5.
why: Match to the options.

Where it goes wrong

- Saying 'at least 0' by analogy with even degrees like $x^2 + 1$, which genuinely have none.
- Saying 'exactly 5', which counts complex roots or ignores repeats.

Takeaway Odd degree $\Rightarrow$ at least one real root; degree $n \Rightarrow$ at most n real roots.

Question 8 GRA-12 · Graphs of functions · spec MM8.1

Which of the following functions is undefined at exactly two real values of x ?

Answer: E

$x^2 - 4 = 0$ at $x = \pm 2$: two values. $x^2 + 4$ is never zero; $\frac{x}{x^2}$ and $\frac{1}{(x-2)^2}$ are undefined at one value; $\sqrt{x-2}$ is undefined for all $x < 2$.

Full method - how and why

'Undefined' means division by zero (or a negative under a square root). Count the offending values for each option.

- 1 $\frac{1}{x^2-4}$: the denominator vanishes at $x = 2$ and $x = -2$ - **two** values.
why: Difference of two squares gives two distinct roots.
- 2 $\frac{1}{x^2+4}$: the denominator is always at least 4, so it is never zero - defined everywhere.
why: A sum of a square and a positive constant is never zero.
- 3 $\frac{x}{x^2}$ and $\frac{1}{(x-2)^2}$: each has a single problem value (0 and 2 respectively).
why: A repeated factor still vanishes at only one point.
- 4 $\sqrt{x-2}$: undefined for **all** $x < 2$ - infinitely many values, not two.
why: Square roots need a non-negative argument.

Where it goes wrong

- Counting $(x-2)^2$ as two values because of the power.
- Thinking $x^2 + 4$ has roots ± 2 .

Takeaway Count the distinct values that break the expression: repeated factors give one, sums of squares give none.

Question 9 GRA-16 · Graphs of functions · spec MM8.7

The line $y = mx$ meets the curve $y = x^3 - x$ at three distinct points. Find the complete set of values of m .

Answer: E

$$x^3 - x - mx = x(x^2 - (1 + m)) = 0: \text{three distinct roots iff } 1 + m > 0, \text{ i.e. } m > -1.$$

Full method - how and why

Intersections of a line through the origin with a cubic. Factor out the common x and use the discriminant on what remains.

- 1 Set them equal: $mx = x^3 - x$, so $x^3 - x - mx = 0$.
why: Bring everything to one side.
- 2 Factor out x : $x(x^2 - (1 + m)) = 0$.
why: $x = 0$ is always a solution, because both graphs pass through the origin.
- 3 The other roots satisfy $x^2 = 1 + m$, which has two distinct non-zero solutions exactly when $1 + m > 0$.
why: If $1 + m = 0$ the root coincides with $x = 0$; if negative, there are no more real roots.
- 4 So $m > -1$.
why: Three distinct points in total.

Where it goes wrong

- Forgetting the root $x = 0$ and requiring three roots from the quadratic.
- Including $m = -1$, where all three roots coincide at $x = 0$.

Takeaway A line through the origin always meets $y = x^3 - x$ there; factor out x and analyse the quadratic that remains.

Question 10 GRA-04 · Graphs of functions · spec MM8.7

The line $y = x + c$ and the curve $y = \frac{1}{x}$ have no points of intersection. Find the complete set of values of c .

Answer: F

$$x^2 + cx - 1 = 0 \text{ always has discriminant } c^2 + 4 > 0: \text{two intersections for every } c. \text{ So there are no values of } c \text{ with no intersection.}$$

Full method - how and why

A 'find the values' question whose honest answer is 'none'. Substituting shows the discriminant can never be negative.

- 1 Set the expressions equal: $x + c = \frac{1}{x}$.
why: Intersections satisfy both equations.
- 2 Multiply by x (noting $x \neq 0$, since $\frac{1}{x}$ is undefined there): $x^2 + cx - 1 = 0$.
why: Clearing the fraction gives a quadratic whose roots are the intersection x -values.
- 3 Compute the discriminant: $c^2 - 4(1)(-1) = c^2 + 4$.
why: The -1 constant is what makes the discriminant unavoidably positive.
- 4 $c^2 + 4 > 0$ for every real c , so there are always two intersections: no value of c avoids them.
why: Also check the roots are non-zero: their product is $-1 \neq 0$, so neither is the excluded $x = 0$.

Where it goes wrong

- Assuming a line and a hyperbola must sometimes miss, and choosing an interval.
- Sign error making the discriminant $c^2 - 4$, which would give $-2 < c < 2$ - an offered option.

Takeaway Do the algebra even when the answer 'feels' like an interval; $b^2 - 4ac$ with $ac < 0$ is always positive.

Question 11 GRA-07 · Graphs of functions · spec MM8.5

The graph of $y=f(x)$ has $f'(x) > 0$ for $x < 1$, $f'(1) = 0$, $f'(x) > 0$ for $1 < x < 3$, $f'(x) < 0$ for $x > 3$. Which of the following is true?

Answer: C

At $x = 1$ the derivative is zero but does not change sign: a stationary point that is not a turning point (an inflexion).

At $x = 3$ the derivative changes from positive to negative: local maximum.

Full method - how and why

Reading a function's shape from the sign of its derivative. A zero derivative **without** a sign change is an inflexion, not a turning point.

- 1 Before and after $x = 1$: $f' > 0$ on both sides, with $f'(1) = 0$.

why: The gradient dips to zero and comes back positive - the curve flattens momentarily but keeps rising.

- 2 So $x = 1$ is a stationary point that is **not** a turning point: a stationary point of inflexion.

why: Turning points require a change of sign in f' .

- 3 At $x = 3$: $f' > 0$ just before and $f' < 0$ just after, so the curve rises then falls - a local maximum.

why: A $+$ $\rightarrow$ $-$ sign change is exactly a local maximum.

- 4 So: local maximum at $x = 3$, stationary inflexion at $x = 1$.

why: Match to the options.

Where it goes wrong

- Assuming $f'(1) = 0$ must be a maximum or minimum.
- Reading the $+$ $\rightarrow$ $-$ change at $x = 3$ as a minimum.

Takeaway $f' = 0$ with no sign change = inflexion; $+$ $\rightarrow$ $-$ is a maximum, $- \rightarrow +$ is a minimum.

Question 12 GRA-08 · Graphs of functions · spec MM8.7

How many real solutions does the equation $x^3 = 3x + 1$ have?

Answer: D

Sketch $y = x^3 - 3x - 1$: turning points at $x = \pm 1$ with $y(-1) = 1 > 0$ and $y(1) = -3 < 0$. A cubic whose local maximum is positive and local minimum is negative crosses the axis three times.

Full method - how and why

Counting roots of a cubic. Find the turning values and see whether they straddle the axis.

- 1 Rearrange to $f(x) = x^3 - 3x - 1 = 0$.

why: Everything on one side, so the question becomes 'how many roots?'.

- 2 Differentiate: $f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1)$, so the turning points are at $x = \pm 1$.

why: A cubic with two turning points can have one, two or three roots.

- 3 Evaluate: $f(-1) = -1 + 3 - 1 = 1 > 0$ (local maximum) and $f(1) = 1 - 3 - 1 = -3 < 0$ (local minimum).

why: The **values** at the turning points decide the count.

- 4 The local maximum is above the axis and the local minimum below, so the curve crosses three times.

why: It comes up from $-\infty$, crosses, comes down past the axis, crosses, then rises again.

Where it goes wrong

- Trying to factorise - this cubic has no rational roots, so factorising wastes time.
- Answering 1 because 'a cubic always has at least one root' without checking the turning values.

Takeaway Cubic root count: find the turning values; three roots iff the local max and min lie on opposite sides of the axis.